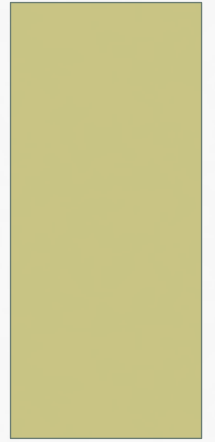


FINISHING AND POLISHING OF RESTORATIONS



Learning outcomes

- List the objectives of finishing and polishing of restorations
- Compare different hand and rotary instruments used for finishing and polishing of restorations.

Introduction

The main objective of an restorative dentist is to restore the individual tooth to its form and functions along with imparting pleasing esthetics



FINISHING AND POLISHING

Finishing and polishing is considered as the last step in the restorative process.

However it should be regarded as one of the most important allowing the actual natural look like with the normal adjacent teeth.



WHY FINISHING AND POLISHING IS REQUIRED

Improper finishing/polishing of dental restorations can result in

- Excessive plaque accumulation
 - Gingival irritation
 - Increased surface staining
 - Poor esthetics of restored teeth
-
- All these could lead to possible recurrent caries and periodontal problems.

DEFINITION



Finishing is described as the transformation of an object from a rough to a refined form.

This finishing process involves removing marginal irregularities, defining anatomic contours, and smoothing away surface roughness of a restoration .

Polishing is described as the production of a shiny mirror like surface, which reflects light similar to natural enamel.



OBJECTIVES OF FINISHING AND POLISHING

1. Oral health: A well contoured finished and polished restoration will have less accumulation of food debris and pathogenic bacteria.

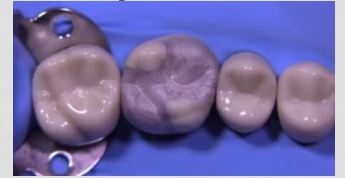
Smoother restoration - much easier to maintain oral hygiene

2. Oral function: Oral function is enhanced with a well polished restoration, because food glides more freely over occlusal and embrasure surfaces during mastication.

3. Esthetics : Aesthetic demands

FINISHING AND POLISHING OF AMALGAM RESTORATIONS

- The final finish of the restoration should not be done until the amalgam is fully set, because crystallization is not complete and compressive strength is not completely attained



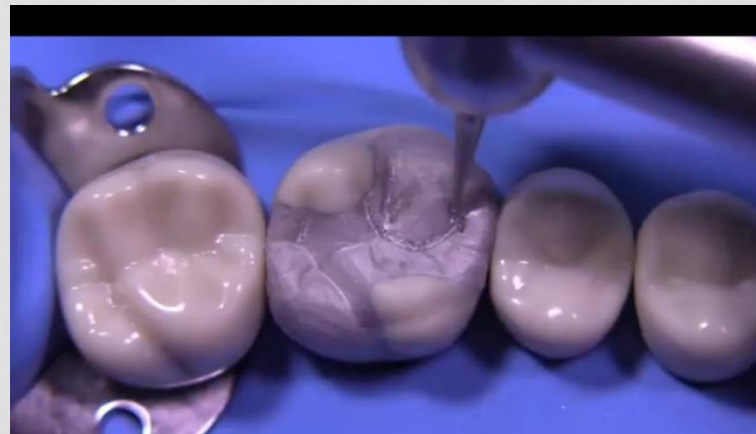
- It should be delayed for 24 hours after condensation .

There is a wide variety of instruments available for finishing, and polishing amalgam restorations.



Finishing burs:

- Finishing burs differ from cutting burs in that finishing bur blades are finer, their sizes smaller, and their number of blades more than cutting burs.



NUMBER OF BLADES IN A BUR

- The number of blades on a bur is usually EVEN

The blades on a

Cutting bur are usually 6 to 10

Finishing bur are usually 12 to 40.



Fluted carbide burs



- After finishing is done, the cavosurface margins should be re-evaluated with an explorer tine



- Polishing is a procedure that makes amalgam restorations look their best.
- Polishing can be carried out in a number of different ways.

- The traditional method describes polishing with flour of pumice mixed with water, followed by tin oxide powder , applied using a prophylaxis cup in a low-speed handpiece.
- This is a low-cost option for polishing, but it has the disadvantage of splatter from the rotating cup, which can be very messy and unpleasant for the patient.



Amalgam polishing system –

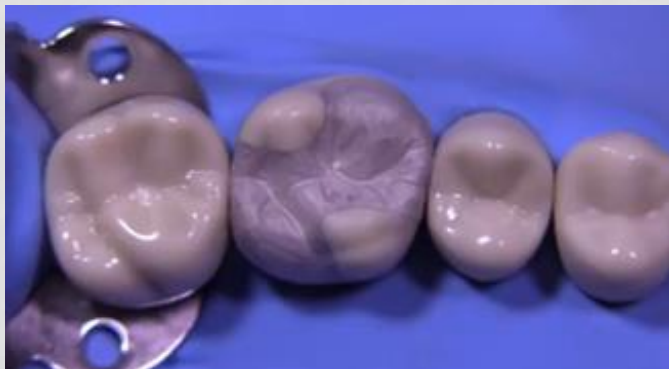
Commercially available rubber polishing system –

- Brownies - Coarse
- Greenies - Fine
- Super Greenies - Extra fine



During polishing, the restoration should be kept moist and only low rotational speeds with light intermittent pressure should be used, to avoid any over heating.

- Excessive heat production (not more than 60°C) during finishing and polishing can permanently damage the pulp



FINISHING AND POLISHING OF COMPOSITE RESTORATIONS

- It is very important step in the completion of the restoration.
- Composite reins have been widely used as an anterior restorative material and are now being used in universal filling material in posterior teeth also.
- Microfilled composite materials can undoubtedly be polished to the highest gloss and are considered to be esthetically best amongst all composites.
- The surface of these microfilled composite restorations is highly smooth and the chances of extrinsic staining are minimal.

- Finishing can be immediately initiated after light curing of composites
- For gross contouring and finishing of the concave like occlusal surface round or oval 12 bladed carbide or finishing diamond can be used .
- Rotary instruments should always be used with a stream of water and little pressure.



YELLOW AND WHITE COATED DIAMOND BURS ARE USED FOR FINISHING

Color coding:

ISO Diamond Grit Identification

N	= Normal/Medium	107-126 μm	ISO code 524	No colour	
SC	= Super Coarse	181 μm	ISO code 544	Black	●
C	= Coarse	151 μm	ISO code 534	Green	●
F	= Fine	40 μm	ISO code 514	Red	●
SF	= Super Fine	20 μm	ISO code 504	Yellow	●
UF	= Ultra Fine	15 μm	ISO code 494	White	○





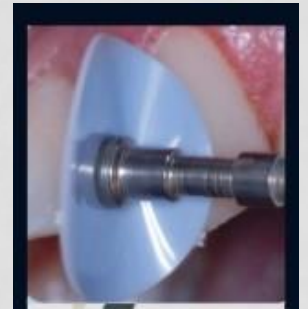
Disks - SofLex disks (3M) or Supersnap disks (Shofu) is available for contouring of facial and palatal surfaces of any teeth

- They work especially well on anterior restorations, such as those involving incisal edges and embrasures, and to a limited extent on posterior composites, especially on interproximal and some buccal-lingual areas.





- These disks are flexible and are produced in several diameters and coated with abrasive like aluminium oxide, silicon carbide.
- Regardless of the type of disk chosen, disks are used sequentially from coarse, medium, fine, superfine generating a smooth surface
- The thin layer of abrasive present on these discs remains effective for a limited period of clinical use, making these discs single-use and disposable



- Proximal and Embrasure Areas.

The proximal surface contours and margins should be assessed visually and tactilely with an explorer and dental floss.

By passing the floss in between the contact one should hear snap sound and “shoe-shined” as it is pulled occlusally.

If the floss catches or frays, additional finishing is indicated.



- Any overhangs at the gingival(proximal)area is removed by Narrow finishing strips may be used to smooth the gingival proximal surface



- Some strips have two different types of abrasives (medium and fine) on opposing ends of the strip, with a small area between where no abrasive is present to allow easy and safe insertion of the strip through the contact area.



FINAL POLISHING OF COMPOSITES

- The traditional method of final polishing is obtained by pumice paste, silica paste. The paste is made by mixing the abrasive like pumice or silica with water and carried over by brushes

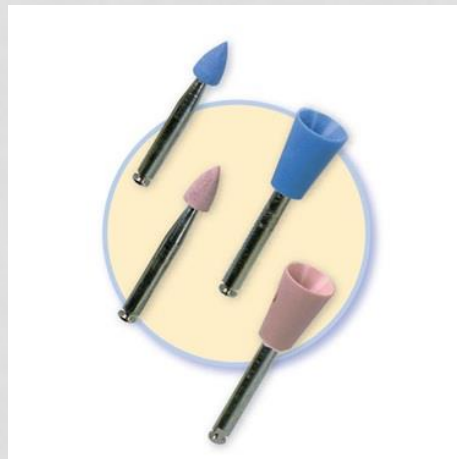


Brushes

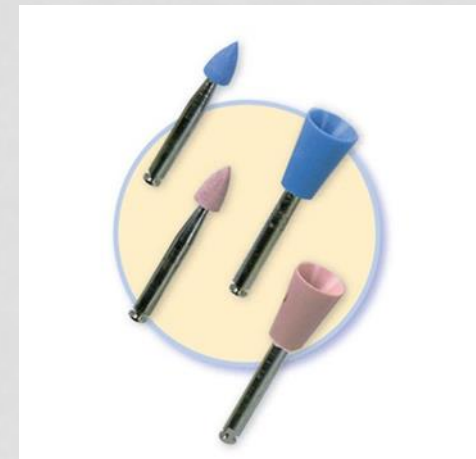


Polishing of the composite

- **Now days** the restorations are polished with appropriate polishing points, cups, spirals.
- These elastomeric, bonded abrasives are usually molded to a “latch-type” mandrel for slow hand pieces.



- The abrasives used within these instruments are usually comprised of silicon carbide, silicon dioxide, aluminum oxide.



FINISHING AND POLISHING OF GLASS IONOMER RESTORATIONS

- The surface of conventional glass ionomer cement is very sensitive to dehydration and water contamination during initial setting phase.
- If the restoration is exposed to air during this period, it loses water and develops cracks.
- Water contamination dissolves the cement and makes restoration weaker
- It is preferable to delay finishing and polishing of the restoration for at least 24 hrs after insertion so that setting is complete in case of conventional self cure GIC

- In case of fast setting cements and light cure glass ionomer cement, finishing can be started as early as 10 minutes after insertion.



- A lubricant such as coca butter or petroleum jelly should be used to protect the restoration from saliva.



PROCEDURE FOR FINISHING

- Excess cement is removed using 12 fluted carbide finishing burs or fine diamond burs are preferred.



- Final smoothing - Soflex disks produced the smoothest surface.



- A lubricant such as coca butter or petroleum jelly should be used to protect the restoration from saliva.



CONCLUSION

- A beautiful statue, even if it is Beautifully sculptured, if left unfinished will attract no spectators.
- Similarly a dental restoration without finishing and polishing will be unesthetic.
- So spend a few more time for finishing and polishing the restoration rather than see all our hard work go in vain.

REFERENCES

- Sturdevant's, 5th edition art and science of operative dentistry
Page no- 521 -522 , 704 -705.
- Craigs 11th edition restorative materials 618 -620

Thank you